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Shunt capacitor bank: Transient issues and analytical solutions

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ABSTRACT

Medium voltage shunt capacitor banks (SCBs) are widely used for improving voltage profile and providing reactive power in electrical networks. Transient oscillations caused by SCBs, e.g., switching and self-excitation phenomena, may damage sensitive equipment in electrical networks. This paper provides an analytical description of the SCB transients and proposes efficient techniques to limit such transients, which include (i) determining the proper size of the SCB to minimum the risk of the self-excitation, (ii) detecting and preventing the SCB self-excitation using a novel protection scheme, and (iii) limiting the SCB switching transients by developing some approaches for medium voltage application. The proposed strategies for limiting SCB transients are verified using comprehensive simulation studies conducted on an actual power system.

1. Introduction

Shunt capacitor banks (SCBs) are widely used for reactive power compensation and bus voltage regulation [1,2]. The cost of an SCB is relatively low compared to the other shunt compensation devices, e.g., SVC and STATCOM and thus SCBs are extensively utilized in power networks [3]. However, the SCB may lead to severe transient oscillations in the power system [4]. For instance, the transients can be caused by SCB energization, back-to-back switching (BTBS), outrush current, and self-excitation [1–8].

The overvoltage and overcurrent transients resulted by SCB switching can adversely affect its own components as well as the equipment connected to the same system [9]. According to the IEEE standard 1036 [5,6], SCB should be able to tolerate voltages and currents in excess of their ratings, i.e.,

- 110% of the rated rms voltage;
- 120% of the rated peak voltage;
- 135% of the rated rms current;
- 135% of the rated reactive power.

The aforementioned continuous limits are intended for contingencies and are not to be used for a nominal design basis. In fact, the minimum lifetime of an SCB is about 100,000 h. Therefore, if other factors do not accelerate the insulation aging, an SCB is expected to operate for the aforementioned time period. Operational incidents such as repeated energization or overvoltages can decrease this lifetime [7–10]. Meanwhile, the maximum time limits for the overvoltages imposed on the individual elements in an SCB are determined in the IEEE Standard [5]. Overall, the SCB transients would originate from:

(1) energization of an SCB;
(2) Back-to-back switching (BTBS), i.e., switching of a second capacitor bank on the same bus in the presence of an already energized bank;
(3) Outrush current, i.e., discharging of an SCB into a short-circuit fault in the upstream system. It should be noted that outrush current amplitude can reach 300 times the rated current.
(4) Self-excitation, i.e., resonance between induction motors and SCB when the system is disconnected from the upstream network.

SCB energization can cause considerable overcurrent and overvoltage transients. The SCB draws a large amount of high-frequency oscillatory current from the network and thus results in a transient overvoltage on the corresponding bus [12–14]. The transients can intensify when there are more SCBs in-service [5]. In addition, when a short circuit fault occurs close to an SCB, it is discharged into the fault location causing a large current flowing out of the SCB [15,16]. The magnitude and frequency of this current depend on the inductance between the SCB, the fault location, and the SCB voltage at fault instant. This current may result in severe stresses on the SCB, the circuit breaker of the faulty feeder, and the corresponding current transformers [17–19].

Although occurrence of some SCB transients are well published, analytical–based defense strategies have not been scrutinized. This issue is highly important for self-excitation phenomenon and thus this paper is more focused on this transient. This paper proposes novel

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methods to mitigate or prevent SCB transients. For this purpose, (i) an analytical method is introduced to determine maximum permissible amount of the SCB rated power to prevent self-excitation phenomenon, (ii) a protective scheme is proposed to resolve the SCB transients from damaging the SCB components and equipment, and (iii) size of the inductor in series with the SCB is determined to reduce the impact of switching transients on the network.

The rest of this paper is organized as follows. Section 2 analytically considers SCB transients. Section 3 describes several simulation studies to reflect SCB transients. Section 4 introduces several proposed techniques and the corresponded requirements to mitigate the SCB transients. Finally, Section 5 concludes the paper.

2. Analysis of SCB transients

SCBs are highly sensitive to the magnitudes and the time durations of overvoltages. For instance, an SCB should not be exposed to 1.7 pu voltage for longer than 1 s [6]. For higher overvoltages the mentioned time would be gradually decreased. Besides, overvoltage occurrence should be limited to a certain number to prevent damages [7]. This section investigates various SCB transients which can cause transient overvoltage or overcurrent stress.

Based on the obtained experience in an actual power system which is studied in this paper, SCB failures occurred during some transients as shown in Fig. 1. After scrutinizing the faulty SCBs, it is found that the failure was resulted by a dielectric breakdown in the SCB insulation due to self-excitation. Indeed, overvoltage was found as the main reason of this accident. On the other hand, utility power grid consumers especially electronic devices are sensitive to the overvoltages caused by the SCB transients. According to [11], electronic devices can tolerate only 1.2 pu for just 0.5 s. These constraints should be taken into consideration before installing SCBs in power systems.

2.1. SCB energization

Fig. 2 shows the Thevenin equivalent circuit of the upstream network where the SCB is switched-in at t = 0 s. In practice, the equivalent resistance is less than the inductance and thus it can be neglected [9]. The natural frequency of the circuit depends on the capacitance C and the equivalent inductance L. Based on the electrical circuit theory, the SCB current has two components:

(1) The steady-state (power frequency sinusoidal) current
(2) The transient current caused by the switching action

The latter is oscillatory due to resonance of the capacitor with the network inductance. The transient component of the SCB current can be calculated as follows [9]:

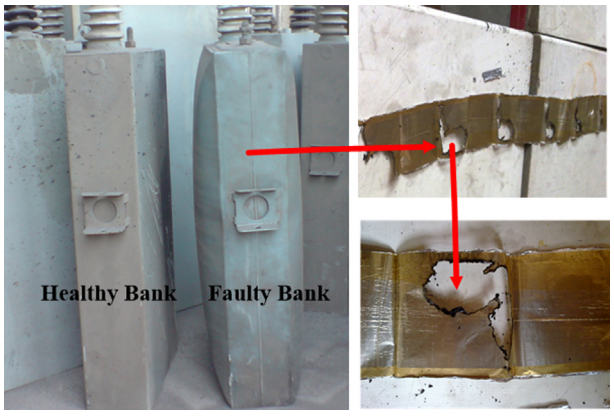


Fig. 1. SCB failure due to transients.

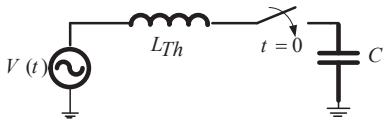


Fig. 2. Simplified equivalent circuit representing the SCB energization.

$$i(t) = \frac{V_0}{Z_0}\sin\left(\frac{1}{\sqrt{L_{Th}C}}t\right) \quad (1)$$

where V_0 is the $V(t)$ magnitude at the switching instant and

$$Z_0 = \sqrt{\frac{L_{Th}}{C}} \quad (2)$$

In addition, the SCB terminal voltage corresponded to the first moments of the energization is approximately:

$$V_c(t) = V_0\left(1 - \cos\left(\frac{1}{\sqrt{L_{Th}C}}t\right)\right) \quad (3)$$

The details of the above calculations are provided in Appendix A. Thus, the magnitude of the switching transient voltage depends on the switching instant and the voltage source magnitude. Eq. (3) indicates that, due to the switching transient voltage, the SCB terminal voltage can rise up to two-times of the network voltage.

2.2. Back-to-Back switching (BTBS)

BTBS involves energizing an SCB on the same bus when another energized SCB is present [13,14], as shown in Fig. 3. In this case, a higher frequency transient occurs as the previously energized capacitor shares its charge with the newly energized capacitor and thus the previously energized capacitor pushes current into the newly capacitor. In fact, the BTBS inrush (transient) current has two components:

(i) the first component is provided by the upstream grid, and is limited by the network impedance;
(ii) the second component is caused by the discharge of other energized SCBs, and can be more severe.

As shown in Fig. 3, before C_2 is connected to the circuit, two equivalent voltage sources can be considered: The Thevenin equivalent voltage $V(t)$ and the initial voltage of C_1. By applying the superposition principle, the transient current injected to the C_2 can be calculated at the energizing instant.

To do this, two circuits are considered: the first one corresponds to the time when the initial voltage of C_1 is assumed to be zero (it is referred as $I_{C2\text{-}1}$). For the second one, the Thevenin voltage is assumed to be zero and the initial voltage of C_1 is equal to V_1 (it is referred as $I_{C2\text{-}2}$). The total transient current of C_2 is the sum of the two aforementioned currents. Based on Fig. 3, L12 is the inductance between C_1 and C_2. By neglecting L_{12} (assumed that C_1 and C_2 are connected to the same bus), the inrush current of C_2 which is supplied by the upstream network, based on (1), is:

$$\begin{cases} I_{Th} = \frac{V_o}{Z_o}\sin\left(\frac{1}{\sqrt{L_{Th}(C_1+C_2)}}t\right),\ Z_o = \sqrt{\frac{L_{Th}}{C_1+C_2}} \\ I_{C2-1} = \frac{C_2}{C_1+C_2}I_{Th} \end{cases} \quad (4)$$

where I_{Th} is the current injected by upstream network and $I_{C2\text{-}1}$ is C_2

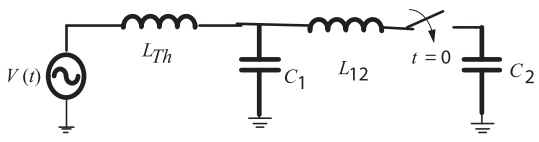


Fig. 3. Simplified equivalent circuit representing BTBS of SCB.

current at connecting instant. In the latter case, the network Thevenin voltage is assumed to be zero and the initial voltage of C_1 is equal to V_1. Thus, C_2 inrush current due to the initial voltage of the C_1 ($I_{C2\text{-}2}$) is approximately [9]:

$$\begin{cases} I_{C_{2-2}} = \frac{V_1}{Z_o}\sin\left(\frac{1}{\sqrt{L_{Th}(C_{eq})}}t\right) \\ Z_o = \sqrt{\frac{L_{12}}{C_{eq}}},\ C_{eq} = \frac{C_1 \times C_2}{C_1 + C_2} \end{cases} \quad (5)$$

The C_2 inrush current is the sum of the two components $I_{C2\text{-}1}$ and $I_{C2\text{-}2}$. In order to limit the C_2 inrush current due to the initial voltage of the C_1, based on (5), it is necessary to increase L_{12}. Therefore, an inductor in series with the SCB is necessary to limit this current.

2.3. Outrush current

When a short circuit fault occurs close to an SCB, it is discharged into the fault causing a large current flowing out of the SCB which is referred to as the outrush current [15,16]. The magnitude and frequency of this current depend on the inductance between the SCB and the fault location. Thus, the peak value of the outrush current is determined by the capacitor initial voltage and the impedance between the capacitor and the fault point. The outrush current can be very severe for a small inductance. This may result in severe stresses on the SCB, the circuit breaker of the faulty feeder, and the corresponding current transformers [17–19].

2.4. Self-excitation of induction motors

One of the most prevalent applications of SCBs is compensation of the reactive power consumed by induction motors [12]. Starting large medium-voltage motors can result in excessive voltage drop in weak power systems. In such a case, the self-excitation phenomenon might occur when an SCB and an induction motor are disconnected from the network, i.e., islanded [12]. In this scenario, the motor temporarily operates as an induction generator. Although the motor has no connection to the network, it temporarily continues to rotate due to the inertia of the rotor and its mechanical load. In this case, the SCB is charged by an induction generator and thus its terminal voltage can increase significantly, which results in damaging the insulation of the equipment connected to the isolated network [18,20]. The equivalent circuit of an induction motor and its SCB is shown in Fig. 4. To simplify the analysis, the Thevenin parameters of the motor equivalent circuit is $R_{eq} + jL_{eq}$ where both of the components are functions of the motor slip, i.e., s. Based on Fig. 4, the equivalent resistance and inductance can be calculated as [21]

$$Z_{eq} = (R_s + j\omega L_s) + (j\omega L_m \| (\frac{R_r}{s} + j\omega L_r)) \quad (6)$$

By considering $Z_{eq} = R_{eq} + jL_{eq}$ and after some simplification, the R_{eq} and L_{eq} can be calculated as follows:

$$R_{eq} = R_s + \frac{(\frac{R_r}{s})L_m{}^2}{(\frac{R_r}{s})^2 + (L_r + L_m)^2} \quad (7)$$

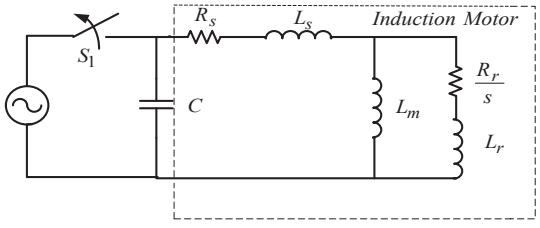


Fig. 4. The equivalent circuit of the induction motor and the shunt capacitor.

$$L_{eq} = L_s + \frac{L_m(L_r{}^2 + L_r L_m + (\frac{R_r}{s})^2)}{(\frac{R_r}{s})^2 + (L_r + L_m)^2} \quad (8)$$

It should be noted that the to calculate R_{eq} and L_{eq}, the motor data provided by the manufacturer is needed. After opening the circuit breaker S_1, the motor and its SCB become islanded from the upstream grid. Under such conditions, the motor terminal voltage can be mathematically calculated by

$$v(t) = e^{-\alpha t}(A\cos(\omega t) + B\sin(\omega t)) \quad (9)$$

$$\alpha = \frac{R_{eq}}{2L_{eq}} \quad (10)$$

$$\omega = \frac{1}{\sqrt{L_{eq}C}} \quad (11)$$

where ω is the circuit resonance angular frequency. When α is negative, the voltage magnitude increases with time, which is referred to as self-excitation. Under such a condition, the SCB, motor and the other equipment in the isolated system may be severely damaged.

3. An actual case study

This section investigates SCB transients in a real-world case study, i.e., the power system of Khousestan steel company in Iran, which is a large-size industrial plant including several SCBs. A simplified circuit diagram of a portion of the aforementioned power system is shown in Fig. 5. The nominal system frequency is 50 Hz. The 40 MVA substation transformer has a voltage ratio of 33/6.6 kV. The transformer is connected to three buses A1, A2, and A3. Each bus consists of a 4.8 Mvar SCB and seven induction motors. The buses A1 and A3 also supply the local static loads that are modeled as series R-L impedances. The L_{Th} of the system from view point of 6.6 kV side is about 0.45 mH. This study system is modeled in the PSCAD/EMTDC software. The specifications of the studied motors are given in the Appendix B.

3.1. Energizing an isolated SCB

This part investigates the energization of an isolated SCB in the study system of Fig. 5. In this case, it is assumed that the SCBs of buses A1 and A3 are disconnected, and the SCB of the bus A2 is assumed to be energized. Fig. 6 shows the SCB voltage and current waveforms under such a condition. In this case, the SCB voltage exceeds 1.9 pu which may result in damages to the network equipment. The SCB steady-state current is about 500 A while the switching peak current is about 4.7 kA. It should be noted that this value is still acceptable and complies with the standard recommendations [5] and the technical specifications of the capacitor manufacturers, due to the short duration of the overcurrent stress [6]. It should be noted that the inrush current can be

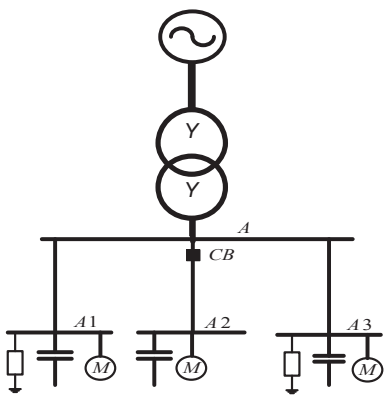


Fig. 5. Single-line diagram of a part of the steel company system.

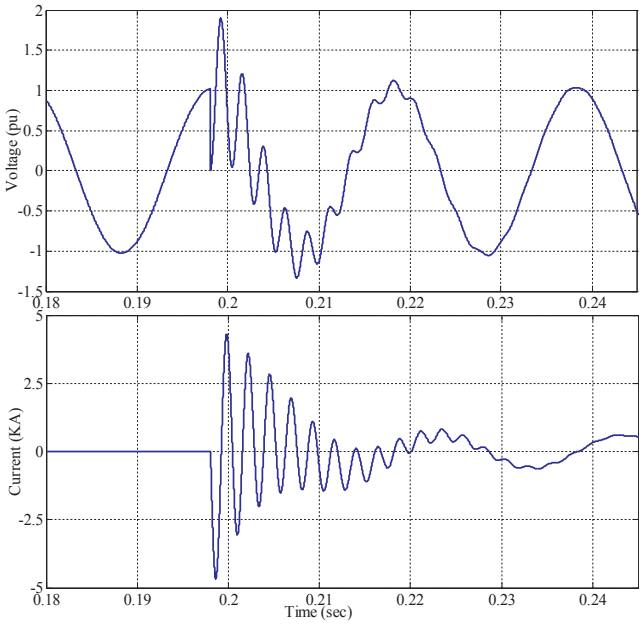


Fig. 6. Voltage and current of bus A2 during energization in the absence of other SCBs.

calculated based on (1). For the system under studied, based on the $L_{Th} = 0.45$ mH and $C = 420$ μF values, peak of the SCB current using (1) is about 4.9 kA and this error is due to neglecting the network resistance.

3.2. Back-to-back switching (BTBS)

When some SCBs are in-service, they reduce the bus voltage variations caused by energization of a new SCB. However, the inrush current of the energized SCB is significantly more intense in this case, due to the contribution of other capacitors. Fig. 7 shows the voltage and current waveforms of the capacitor on bus A2 during the energization of SCB on A2, when the SCBs of the buses A1 and A3 are in-service. The simulation studies show that voltage of the bus A2 reaches 1.2 pu, which is

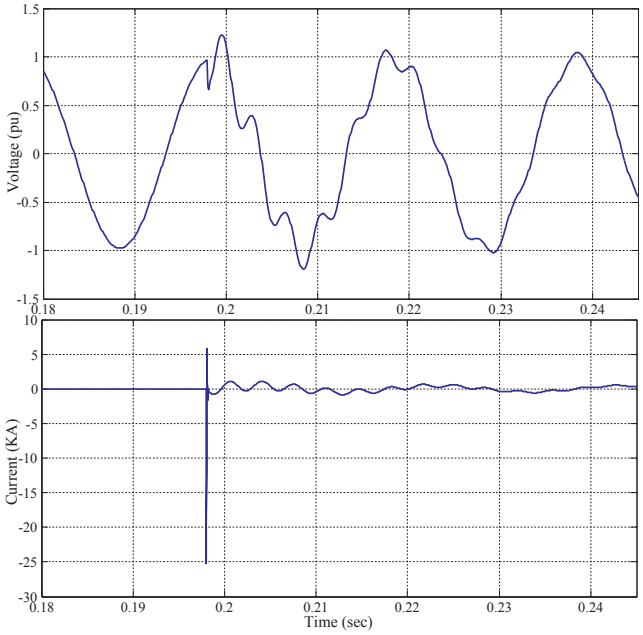


Fig. 7. Voltage of the bus A2 and current of its SCB during BTBS.

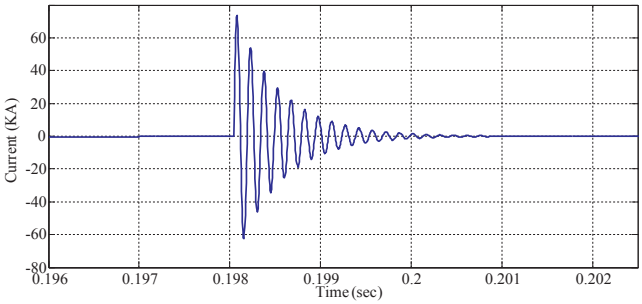


Fig. 8. Current of the SCB connected to the bus A2 during outrush.

considerably lower than the case where the other SCBs are disconnected. However, the SCB current of A2 reaches 25 kA, which is substantially larger than the case where the other SCBs are disconnected (i.e., 4.7 kA). This large energization current

passes through the SCB and the upstream network equipment, and thus can cause serious damage to the equipment. Therefore, it is necessary to limit the magnitude of this transient current.

3.3. Outrush current

To investigate the SCB outrush current, a three-phase short-circuit is applied to the bus A when all three SCBs are in-service. Fig. 8 shows the current of the SCB connected to the bus A2, where the peak current reaches 75 kA. This can seriously damage the SCBs and the other network equipment. Therefore, the outrush current should be limited as well.

3.4. Induction motor self-excitation

To study the self-excitation phenomenon, it is assumed that the feeder connected to the bus A2 experiences an interruption at $t = 1$ s, and thus the bus A2 becomes islanded from the upstream grid. In this scenario, just three motors are in-service. Under this condition, the SCBs connected to the bus A2 as well as the motor can be affected by the stresses associated with the self-excitation phenomenon. As shown in Fig. 9, the voltage of the bus A2 increases and this overvoltage condition lasts for about 2 s. In fact, as the motor coasts down and the frequency of its air-gap flux declines, voltages in the isolated system build up and subsequently start to decline. The study results show that if size of SCB increases, the magnitude of the transient overvoltage becomes significantly higher. In addition, the results demonstrate that the overvoltages can even last for 5 s. Such a large voltage stress can lead to destructive damages to the system equipment. It should be noted that the rated power of the SCB is usually less than the maximum consumed reactive power of the loads connected to the same bus. However, when the SCBs are not controllable using a power factor regulator system, reactive power overcompensation can occur under some operation conditions, i.e., when some of loads are out of service.

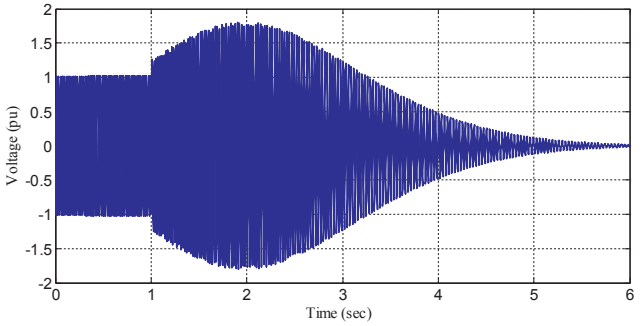


Fig. 9. Bus A2 voltage waveform during the self-excitation event.

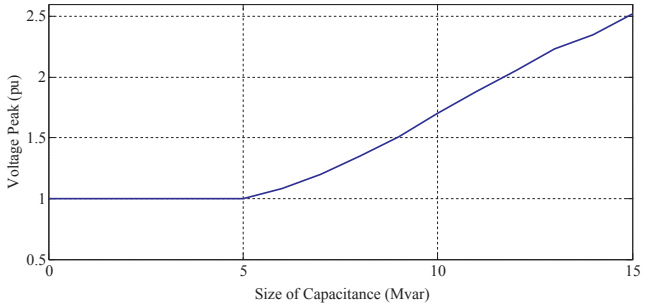


Fig. 10. Voltage magnitude of Bus A2 after the islanding and self-excitation mode.

For instance, in the system under study, four motors connected to the bus A2 may sometimes be out of service while the SCB is connected to the bus. In this case, after disconnection of the bus A2 from the grid, the islanded system voltage would be greater than 1.5 pu for more than 2 s. Fig. 10 shows dependency of the SCB rated power on the voltage amplitude experienced under self-excitation conditions. As illustrated in Fig. 10, when the reactive power consumed by the motor is equal to 4 Mvar and the SCB capacity is 5.1 Mvar, the isolated bus does not experience an overvoltage stress. However, when the reactive power compensation is more than 2 times of the load reactive power, the bus voltage exceeds 1.5 pu, which may damage both SCB and motors. Therefore, the self-excitation phenomenon takes place in over-compensated systems and thus this is theoretically analyzed in the following section.

4. Proposed approaches to mitigate SCB transients

In this section a two-step solution is proposed to mitigate the self-excitation problem. At first, the capacity of SCB is determined in a way that no destructive overvoltage stress occurs after the system is disconnected from the grid. As the second step, a protection scheme is proposed to quickly detect self-excitation event and disconnect the SCB before destructive overvoltages take place. In addition, effective techniques are proposed to limit the excessive inrush/outrush current and overvoltages during the SCB switching.

4.1. Self-excitation prevention by limiting in-service SCBs

To prevent self-excitation, at first theoretical analysis is developed to provide more realization from this phenomenon and then the proper size of SCB based on the obtained results is determined.

4.1.1. Self-excitation phenomenon theoretical analysis

According to (9), motor terminal voltage theoretically increases after being disconnected from the upstream network, when $\alpha = R_{eq}/2L_{eq}$ becomes negative. Since based on (8) L_{eq} always has a positive value, the self-excitation can happen when R_{eq} becomes negative. Based on (7), this can happen due to a negative value of the induction motor slip, which is defined as:

$$s = \frac{\omega_s - \omega_m}{\omega_s} \tag{12}$$

where ω_s and ω_m represent the synchronous electrical angular velocity and the rotor mechanical angular velocity, respectively. Once the motor is connected to the network, $\omega_s = 2\pi f_s$, where f_s is the system nominal frequency. Since ω_m is less than ω_s in a grid-connected induction motor, the slip s is positive.

When an islanded event, the electrical frequency f_r is equal to the resonance frequency of the equivalent L-C circuit, as illustrated in Fig. 5. On the other hand, the out-of-service motor mechanical speed ω_m gradually reduces. In this case, the machine slip is determined as

$$s = \frac{\omega_r - \omega_m}{\omega_r} \tag{13}$$

$$\omega_r = \frac{1}{\sqrt{L_{eq}C}} \tag{14}$$

If the resonance angular frequency ω_r becomes less than ω_m, the slip s would be negative and thus the machine operates as an induction generator which is called self-excitation phenomenon.

4.1.2. Proper SCB sizing to prevent self-excitation phenomenon

To prevent the generator-mode operation of the induction machine after its disconnection from the upstream system, its associated slip should remain positive. This means, based on (13), ω_r should be more than ω_m. Thus:

$$\omega_r = \frac{1}{\sqrt{L_{eq}C}} > \omega_m \tag{15}$$

Based on (15), the condition for preventing self-excitation is:

$$C < \frac{1}{L_{eq} \times {\omega_m}^2} \tag{16}$$

Based on Fig. 4, since L_m is significantly larger than both L_r and R_r/s, (8) can be simplified as

$$L_{eq} \simeq L_s + L_m \tag{17}$$

where L_m is the magnetic inductance of the induction machine, and L_s is the stator winding leakage inductance. Mechanical motor speed gradually decreases to become zero after disconnection from the grid. Thus, to determine C in (16), as the worst condition, ω_m can be considered as the motor speed before interruption instant. Meanwhile, due to the small slip value during the motor normal operation, just for simplicity and without losing generality, it is assumed that $\omega_m = \omega_s = 2\pi \times 50$ rad/s.

It should be noted that (16) does not provide any information about the magnitude of the overvoltage at the machine terminal after disconnection from the grid. Simulation studies and experimental observations show that deviation from the constraint expressed in (16) does not lead to serious overvoltages at the terminals of the interrupted machine. Hence, the coefficient K is used to determine the critical SCB. Thus, (16) can be expressed as

$$C < \frac{K}{(L_m + L_s) \times {\omega_s}^2} \tag{18}$$

The results of the sensitivity analysis performed using extensive simulation studies indicate that for $K = 1.5$, the overvoltage after the islanding is limited to 1.1 pu, which is tolerable for SCBs, induction machines, and other network equipment. It should be noted that where multiple induction motors are connected to a bus, similar to bus A2 in the system under studied which feeds 7 induction motors, the equivalent inductances of the parallel-connected motors should be considered. For instance, according to (18), by calculating equivalent inductance of the motors, the maximum size of the SCB that can be used in A2 is equal to 400 μF, which at the nominal voltage of 6.6 kV is equal to 5.6 Mvar. If some static loads are also connected to the bus, since these loads are connected in parallel with the motors, size of the capacitance would be calculated as follows:

$$C = \frac{K}{(L_{m-eq} || L_{s-eq} || L_{load}) \times {\omega_s}^2} \tag{19}$$

where, L_{Load} is equivalent inductance of the static loads. In addition, $L_{s\text{-}eq}$ and $L_{m\text{-}eq}$ are the equivalent inductance of the motors as well. For instance, seven induction motors are connected to bus A1 along with 3.4 Mvar reactive power of static loads. Based on (19), the maximum permissible SCB to prevent transient overvoltage more than 1.1 pu during a sudden bus A1 de-energizing is obtained equal to 9 Mvar.

The permissible SCB rated power is obtained based on the case that

all motors and static loads are in-service. However, under some operation conditions, some motors and static loads may be out-of-service while all SCBs are in-service and thus this condition has potential to fall in self-excitation. To avoid destructive overvoltages caused by the self-excitation under the mentioned condition, the following actions can be taken:

(1) Limiting the SCB rated power based on the minimum amount of load on the bus. However, this is not usually a practical approach since it contradicts with other essential requirements such as achieving desired power-factor under the maximum load conditions.
(2) Utilizing a power-factor regulator to disconnect a part of the SCB depending on the amount of loads connected to the bus. This approach necessitates dividing the SCB into several smaller sets, each of which should be equipped with its own circuit breaker. This approach suffers from the following issues:
   – Utilizing circuit breaker for each SCB increases the corresponding cost.
   – Switching the capacitor sets considerably increases the SCB failure rate due to SCB switching transients.
(3) Applying a protective scheme to disconnect the SCBs when self-excitation is detected and before considerable overvoltage appears. This approach, which is discussed in the next section, allows the system to operate even with a lead power factor.

4.2. Proposed self-excitation protection scheme

In industrial systems including the study system of Fig. 5, over-compensation is a common practice under light load conditions especially during motor starting and thus this may lead to self-excitation. Therefore, two protection strategies are proposed in the following subsection to prevent unacceptable overvoltage stresses.

4.2.1. Proposed event-based approach

Preventing self-excitation in an isolated bus feeding induction motors and SCBs can be realized by real-time monitoring the status of the upstream circuit breakers. If an islanding event is detected, a trip signal would be issued to all or a part of SCBs. Although this method can practically be implemented, receiving real-time signals from various locations, i.e., status of circuit breakers, and its verification are serious issues. In addition, unwanted SCB disconnection caused by undesirable signals due to noise conditions may lead to congestions in the system, such as overload or undervolatge.

4.2.2. Proposed response-based approach

To prevent occurrence of self-excitation, the natural frequency of the system should be greater than the nominal frequency, and thus the following condition should be met:

$$\frac{K}{\omega_s{}^2 L_{eq}} > C \Rightarrow \frac{KV^2}{L_{eq}\omega_s} > V^2 C\omega_s \Rightarrow KQ_{eq} > Q_c \tag{20}$$

where for each bus, Q_{eq} is inductive reactive power of both static and dynamic loads and Q_c is the SCB reactive power. As explained earlier, reactive power compensation more than K times the load reactive power has self-excitation potential and thus transient overvoltage greater than 1.1 per-unit which is most likely to be appeared. Therefore, based on measuring the power-factor of each bus, if the following constraint based on (20) is met, the SCB should be tripped to prevent self-excitation:

$$KQ_{eq} - Q_C < 0 \tag{21}$$

Sometimes, in order to reduce motor starting time, it is necessary to connect more SCBs to increase the bus voltage and compensate voltage drop due to the starting current. To decrease the starting time of the motor, the acceleration torque and thus the electrical torque should be increased during starting process. To increase electrical torque, the terminal voltage should be increased. It should be noted that electrical torque of the motor is proportional to the square terminal voltage [21]. By increasing the SCB, the bus voltage value would be increased and thus decrease the starting time process.

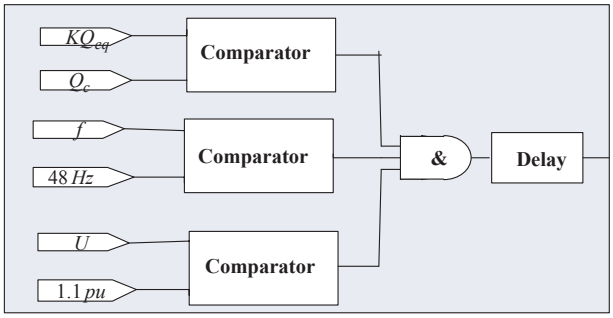


Fig. 11. Proposed self-excitation protection scheme.

In this situation, the proposed protection scheme may have mal-operation and thus should be disabled during motor starting. To prevent this mal-operation, it is essential to use additional protection function units which are an under-frequency and under voltage protection relays:

– After occurring islanding, the electrical frequency would decrease to zero. The frequency pickup value is set at a lower nominal frequency, say 48 Hz, and a time delay of 100 ms is utilized.
– An overvoltage protection relay with pickup voltage of 1.1 pu and a time delay of 100 ms is utilized to immune the protection scheme against mal-operation as well.

Fig. 11 shows the schematic diagram of the proposed self-excitation protection scheme.

4.2.3. Protective scheme requirements and evaluation

This part briefly describes the requirements of the proposed protection scheme.

4.2.3.1. Requirements for potential transformers (PTs). Since the system frequency during self-excitation is lower than the nominal value, the iron cores of the PTs would be saturated and consequently the bus voltages would be underestimated. Thus, it is recommended to use PTs with large voltage factor of knee-point to postpone saturation, e.g., 1.9 [22].

4.2.3.2. Requirements for voltage calculation. Commercial overvoltage relays operate based on the first harmonic estimation or effective voltage calculation, and in normal operation of the network, in which the system frequency is approximately equal to the nominal value, the voltage obtained from both methods are almost identical. However, after occurrence of self-excitation, the frequency of the system can be different from the nominal value. Therefore, the first harmonic estimation method may cause a large error. Therefore, in order to protect the network against self-excitation, it is necessary to use a relay which utilizes the effective voltage calculation.

To verify performance of the proposed protection scheme, it is assumed that the bus A2 becomes islanded at $t = 0.2$ s. Both mentioned requirements are considered as well. Before this event, only two out of the seven motors are in-service. As shown in Fig. 12, after islanding the bus voltage considerably increases and reaches the maximum value of 3.3 pu. Consequently, the equipment on this bus experience considerable overvoltage that lasts more than 2 s. Since the bus electrical frequency is significantly decreased and also the voltage is increased, the proposed scheme detects self-excitation at $t = 0.33$ s, i.e., less than

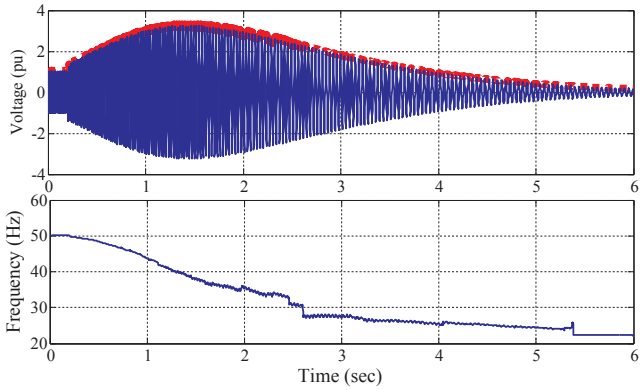


Fig. 12. Behavior of the voltage and frequency estimation during self-excitation.

150 ms after the islanding. Subsequently, these overvoltages are prevented by tripping the SCB.

4.3. Proposed approach to limit switching transients

Basically, there are three approaches to minimize switching transients including energization, BTBS and outrush. The first approach is connecting the SCB at an instant when the system voltage is equal to the SCB voltage. This approach, which can be utilized even for an SCB with trapped charge, requires a circuit breaker that is precisely and independently controllable for each phase. Since the breakers used for the medium voltage capacitors have mainly three-pole (integrated) operating mechanisms, this approach is not practical [12–15]. The second approach is using surge arrester that can be used for limiting overvoltage transients. However, surge arresters may be exposed to high thermal stresses during SCB switching, particularly due to restrike after SCB outage.

The third approach is inserting some impedance, e.g., resistance or inductance, in series with SCB to minimize the transients. The use of pre-insertion resistors is one of the classical methods to minimize the severity of the switching transients by momentarily providing greater losses in the circuit, which leads to reduction of the peak values of the voltage and current transients [12–15]. Utilizing an inductor in series with the SCB is also a commonly used approach to limit the higher-frequency components of the current and voltage transients [14–20]. A method to calculate the proper size of the inductor is discussed in the following section.

4.3.1. Proper series inductor

Increment of the series inductance would reduce the SCB inrush current at the energization instant. For the case that SCB is series with the inductor, the short-time (less than sub-cycle) voltage deviation switching from the nominal value is approximately equal

$$\Delta V = \frac{L_{Th}}{L_{Th} + L_s} \tag{22}$$

The parameter L_s is the series inductance and L_{Th} is the system Thevenin inductance. It should be noted that the resistance of the upstream network is neglected in (22). According to (22), to limit ΔV to 0.5 pu (the absolute bus voltage magnitude equals to 1.5 pu), it is sufficient to select $L_s = L_{Th}$. By increasing L_s, the transient overvoltage and the stress imposed on the equipment would be reduced. It should be mentioned that the inductor results in increasing the normal operating voltage of the SCB. This issue is more important especially for the week system, i.e., large L_{Th}. Therefore, to consider this major issue, another constraint is used as follows:

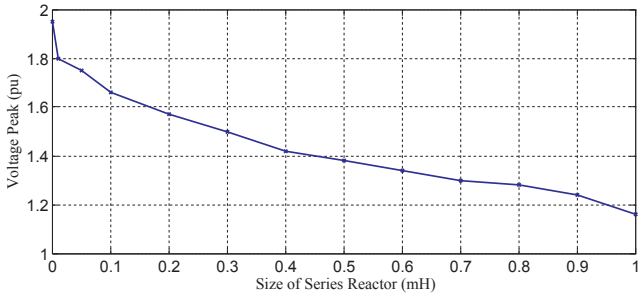


Fig. 13. Effect of the series inductance on the energization overvoltage.

$$V_c = \left|\frac{\frac{1}{j\omega_s C}}{\frac{1}{j\omega_s C} + j\omega_s L_s}\right| \times V_{bus} = \left|\frac{1}{1 - \omega_s{}^2 L_s C}\right| \times V_{bus} < 1.1 \tag{23}$$

where the voltage across the SCB (V_C) should be less than 1.1 pu during normal condition. This value is determined based on IEEE standard 1036 that an SCB should tolerate continuously. In addition, to satisfy (23) for all operational conditions, the bus voltage value (V_{bus}) is selected corresponded to the maximum permissible operational voltage. This value for the studied system is equal to 1.05 pu as well.

The impact of the series inductance on reducing the short-time overvoltage switching on the bus A2 is shown in Fig. 13. Since L_{Th} is equal to 0.45 mH, it is expected that using a series inductor with L_s = 0.45 mH limits the transient bus voltage to 1.5 pu. From Fig. 13, it is clear that the series inductance of 0.45 mH would limit the transient bus voltage magnitude to 1.4 pu, and the 0.1 pu difference between the approximate equation (21) and the simulation result is due to neglecting the network resistance. In addition, as shown in Fig. 14, L_s results in maximum voltage drop of the SCB during normal condition becomes 1.07 pu and thus this voltage is acceptable for the SCB and there is no need to decrease L_s.

It should be noted that the consumption reactive power of the series inductor is about 0.075 Mvar. Therefore, this inductor decreases efficiency of the SCB about 1.56%. However, we accept this issue due to existing practical limitations as mentioned in Section 4.3.

4.3.2. Effect of series inductance on BTBS

BTBS results in considerable inrush current, but as mentioned previously the resulting overvoltage stress is far less than the case where no other SCB is connected to the network. Therefore, the SCB current during BTBS should be considered for extending the lifetime of the system. Assuming n-1 similar SCBs connected to the bus, the current of the nth SCB, due to BTBS, at the switching instant is equal to:

$$I_{BTBS} = \frac{\frac{n-1}{n}V_0}{\sqrt{\frac{L_s}{C}}} \tag{24}$$

where it is assumed that each SCB has a series inductance L_s. According to (24), if L_s equals to the equivalent network inductance, the SCB current during BTBS is approximately similar to the case where no

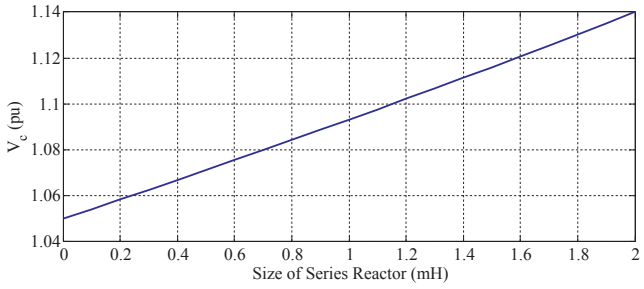


Fig. 14. Effect of the series inductance on SCB voltage during normal condition.

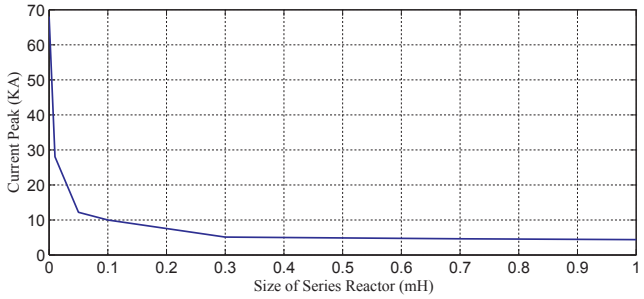


Fig. 15. Effect of the series inductance on the outrush current magnitude.

other SCB is connected and thus the current is smaller than the permissible level.

4.3.3. *Effect of series inductance on the outrush current*

The outrush phenomenon causes large SCB currents. Under a three-phase short-circuit fault at a bus to which SCBs are connected, the current magnitude of each SCB with a series inductor L_s is equal to:

$$I_{outrush} = \frac{V_0}{\sqrt{\frac{L_s}{C}}} \tag{25}$$

where, V_0 is the bus voltage at the fault instant. From (25), it is clear that selecting a series inductance equal to the network equivalent inductance can limit the outrush current to the magnitude corresponded to the SCB energization in the absence of other SCBs. Fig. 15 shows that using a series inductance of 0.45 mH will reduce the peak magnitude of the SCB current from about 70 kA to 5 kA.

5. Conclusions

In this paper, the SCB transient behavior and its impact on the power system have been investigated and explained in details. The size of SCB is analytically determined taking into account the self-excitation phenomenon which imposes considerable transient overvoltage stress on the network and the SCB. Subsequently, a protection scheme is proposed for preventing this phenomenon under light load conditions. The studied results indicate that energization of an SCB, either isolated or in the presence of other SCBs, can impose severe transient overvoltage and overcurrent stresses on the SCBs and the other network components. The study results indicate that series inductances should be also utilized to limit transients in order to prevent damage to the both SCB and system equipment. An analytical approach is introduced to determine the appropriate series inductance value as well.

Appendix A

It is assumed that the SCB is energized at time t = 0 s. Since, *V(t)* source is set to $V_m\cos(\omega_s t)$, thus the *V(t)* magnitude at the switching instant is equal to $V_0 = V_m$. In addition, it is assumed that the SCB voltage before the energization is zero. Based on the electrical circuit theory, there is:

$$\begin{cases} -V(t) + V_L + V_C = 0 \\ V_L = L_{Th}\frac{di_L}{dt} \\ i_C = C\frac{dVc}{dt} \Rightarrow L_{Th}C\frac{d^2Vc}{dt^2} + V_C = V(t) = V_m\cos(\omega_s t) \\ i_L = i_C \\ V_C(0^-) = V_C(0^+) = 0 \end{cases} \tag{A1}$$

Using the Laplace transform, $V_C(t)$ is calculated as follows:

$$V_C(t) = \frac{V_m\cos(\omega_0 t) - V_m\cos(\omega_s t)}{(\frac{\omega_s}{\omega_0})^2 - 1}\,\omega_0 = \frac{1}{\sqrt{L_{Th}C}} \tag{A2}$$

Since, ω_0 (the natural frequency) is considerably more than the ω_s(the system nominal frequency) [9], therefore, the term of $(\frac{\omega_s}{\omega_0})^2$ in (A2) can be approximately neglected and thus $V_C(t)$ is:

$$V_C(t) = V_m\cos(\omega_s t) - V_m\cos(\omega_0 t) \tag{A3}$$

Based on (A3), the SCB voltage has two components: (i) The steady-state; (ii) The transient voltage. To calculate $V_c(t)$ just the first moments of the switching action, corresponded to the time period ofω_0, i.e., $0 < t < \frac{2\pi}{\omega_0}$, it can be assumed that for the mentioned period the $\omega_s t$ is approximately equal to zero and (A3) is written as follows:

$$V_C(t) = V_m - V_m\cos(\omega_0 t) \tag{A4}$$

In addition, the transient current passed through the SCB during energization is:

$$i(t) = C\frac{dV_C}{dt} = \frac{V_m}{Z_0}\sin(\omega_0 t)Z_0 = \sqrt{\frac{L_{Th}}{C}} \tag{A5}$$

It should be noted that since in practice, the equivalent resistance of the upstream network is smaller than the network reactance, without loss of generality and just for simplification, to derive equation (3) (corresponds to the voltage of the SCB after the first moments of the switching), it is assumed that the resistance of the network can be neglected. The peak of the transient voltage by considering resistance would be nearly the same as the pervious results. However, resistance of the network results in damping of the transient voltage and thus by increasing it, this voltage would be faster damped and disappeared after the switching event.

Appendix B

Parameters of A sample 6.6-kV Induction machine

Parameters	Value
Rated Voltage	6.6 kV
Rated Power	2200 kW
Nominal Speed	1440 rpm
L_m	3.37 pu
L_s	0.03 pu
Induction Machine Inertia Time Constant	4.23 s

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